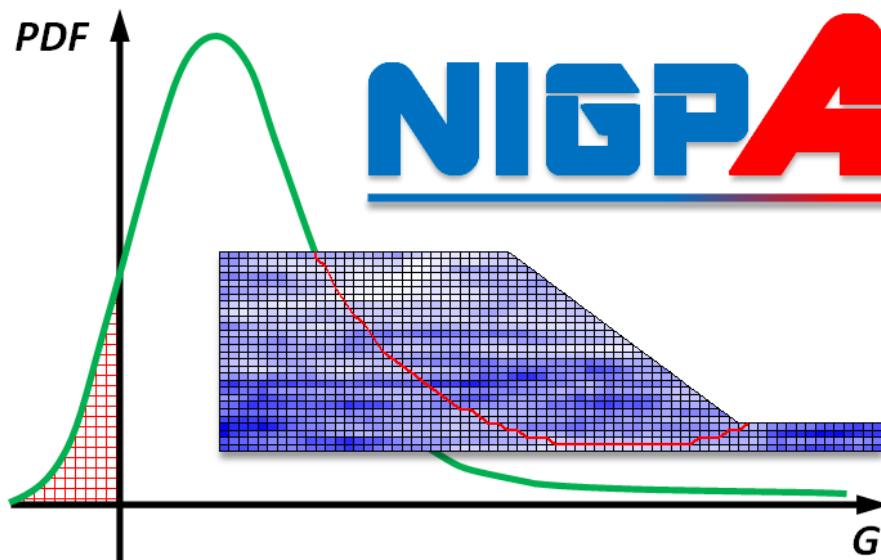




# Introduction to NIGPA

(Version 1.1)

Dian-Qing LI, Te XIAO, Xin LIU, Zi-Jun CAO



Wuhan University  
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# 1 Introduction

NIGPA (Non-Intrusive Geotechnical Probabilistic Aalysis) toolbox (Li et al., 2016a) is a versatile and user-friendly Matlab toolbox that aims to evaluate the reliability of geotechnical structures in a nonintrusive manner. It is deliberately framed for the practical use of geotechnical engineers who are well-trained in deterministic geotechnical analyses but have limited training in probability theory and statistics.

With NIGPA, practitioners only need to develop a deterministic analysis model that they are familiar with using commonly-used commercial software, and to input the estimated statistics and probability distributions of soil properties. After that, NIGPA will simulate soil properties stochastically based on uncertainty modeling, invoke the deterministic model repeatedly through uncertainty propagation, and return reliability analysis results as outputs. These procedures are implemented in a nonintrusive manner, as shown in Figure 1.1 schematically. Although a thorough understanding of probability theory, statistics and reliability algorithms is always advantageous, it is not a prerequisite for engineers to use the NIGPA toolbox. Hopefully, NIGPA can contribute to promoting the application of probabilistic analysis in geotechnical practice.

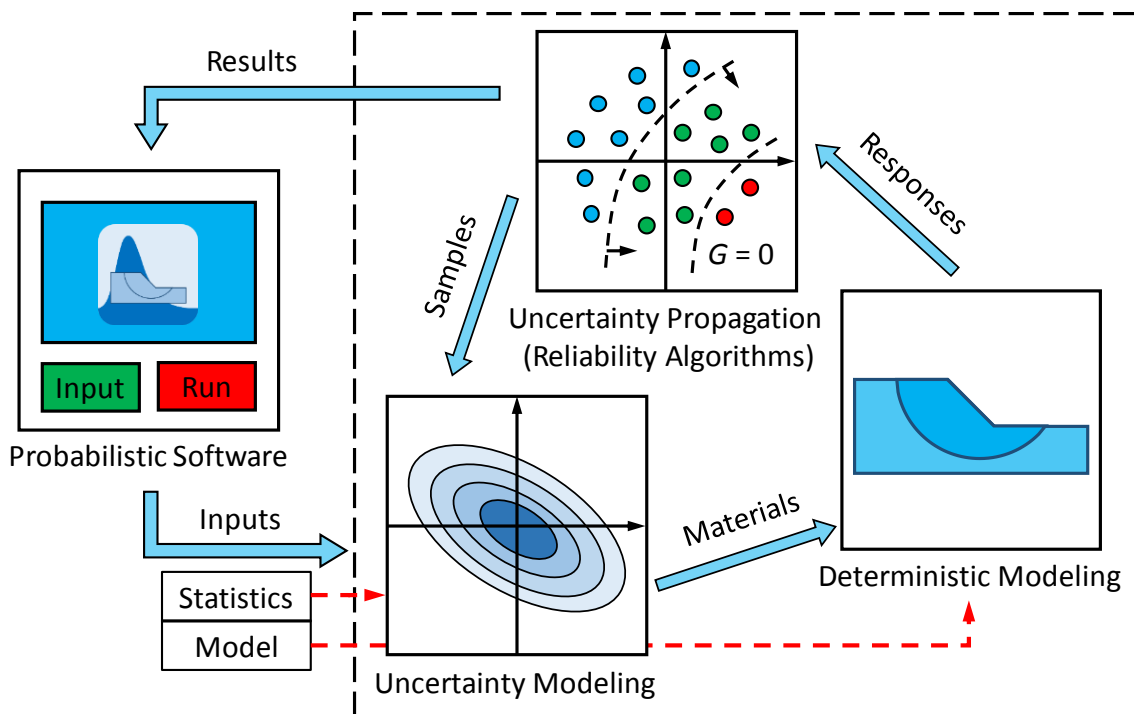


Figure 1.1 Schematic diagram of nonintrusive manner for probabilistic analysis

## **1.1 Requirements**

NIGPA takes advantage of the new graphical user interface development environment available from Matlab R2016a (v9.0) (The MathWorks, Inc., 2016) onwards. The third-party commercial software invoked by NIGPA should be successfully installed by users themselves, such as GeoStudio 2007/2012 (GEO-SLOPE International Ltd., 2016) and Abaqus 6.12/6.13/6.14 (Dassault Systèmes, 2016).

## 2 Theory

### 2.1 Framework

NIGPA consists of four major modules as shown in Figure 2.1, including deterministic modeling, uncertainty modeling, uncertainty propagation, and result visualization. These four modules exactly correspond to four analysis procedures of NIGPA. Among them, the first three modules are key modules of NIGPA and will be introduced in the following sections.

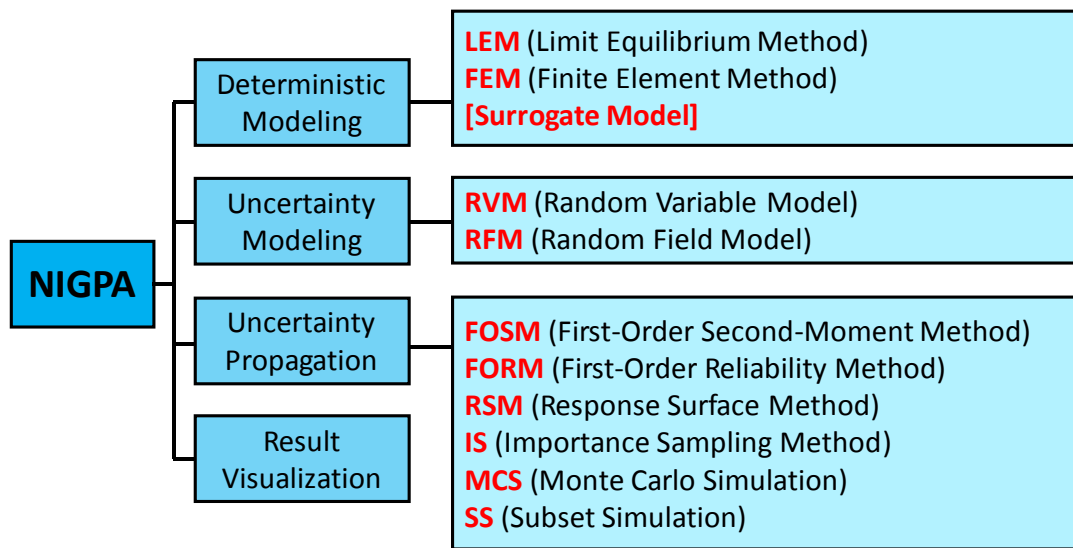


Figure 2.1 Framework of NIGPA

### 2.2 Deterministic Modeling

For geotechnical stability analysis, limit equilibrium method (LEM) and finite element method (FEM) are two commonly-used approaches. Both of them have been successfully applied to geotechnical probabilistic analysis for different geotechnical structures. More details can be referred to Li et al. (2016b). Figure 2.2 demonstrates slope stability analysis models using LEM and FEM for example.

In NIGPA, the deterministic analysis is achieved by third-party commercial software in a nonintrusive manner. The commercial software GeoStudio and Abaqus are invoked for LEM and FEM, respectively. In addition, surrogate model for LEM or FEM (i.e., response surface, as explained later) also can be used in NIGPA to perform deterministic analysis for saving computational costs.

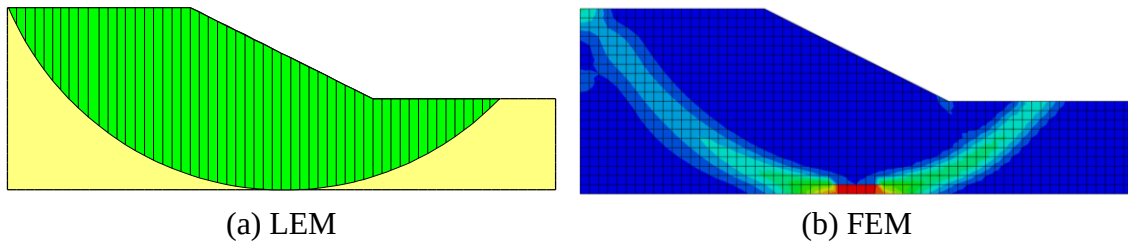


Figure 2.2 LEM and FEM models for slope stability analysis

### 2.2.1 Limit Equilibrium Method (LEM)

LEMs (Duncan and Wright, 2014) are widely used in geotechnical engineering practice since they are conceptually simple and computationally efficient. Different LEMs have different assumptions and equilibrium conditions. Consider the commonly-used ordinary method of slices, simplified Bishop's method and Spencer's method for example. Ordinary method of slices assumes circular slip surface and satisfies moment equilibrium. Simplified Bishop's method assumes circular slip surface and satisfies moment equilibrium and force equilibrium in vertical direction. Spencer's method is available for arbitrary convex slip surface and satisfies moment equilibrium and force equilibrium in both vertical and horizontal directions. No matter which LEM is used, it needs to assume the shape and location of slip surfaces based on engineering judgements and to identify the most dangerous slip surface among a large amount of potential slip surfaces.

### 2.2.2 Finite Element Method (FEM)

FEM can automatically locate the most dangerous slip surface without assumptions on shape and location of slip surfaces. It can also take advantage of soil constitutive behavior and provide deformation information to monitor progressive failure. For geotechnical stability analysis, shear strength reduction technique (Griffiths and Lane, 1999) is usually adopted in FEM to evaluate the safety margin of geotechnical structures, which takes more computational time than LEMs. Besides, since FEM is conceptually complicated, it demands for well-trained users.

## 2.3 Uncertainty Modeling

Inherent variability of soil properties is one of the major uncertainties that significantly affect geotechnical reliability and risk (Phoon and Kulhawy, 1999). NIGPA provides two types of probabilistic models of uncertain soil parameters, including random variable model (RVM) and random field model (RFM), as shown in Figure 2.3. The former ignores the spatial variability of soil properties, while the latter explicitly models it using random field theory (Fenton and Griffiths, 2008).

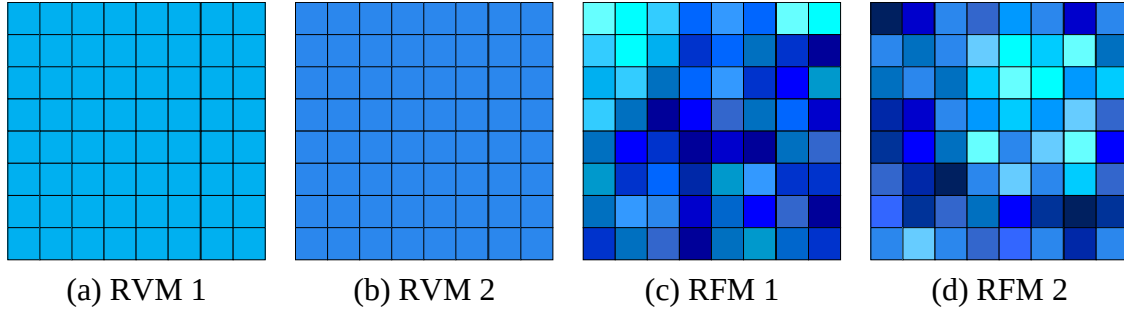


Figure 2.3 Random realizations for RVM and RFM

### 2.3.1 Random Variable Model (RVM)

Normal distribution and lognormal distribution are two commonly-used probability distribution types to describe the inherent variability of soil properties in geotechnical engineering. Only mean and standard deviation are required in their probability density functions, and they can be estimated using samples obtained from field tests or laboratory tests. Compared with normal distribution, lognormal distribution does not allow negative values.

### 2.3.2 Random Field Model (RFM)

In the context of RFM, the spatial correlation should be specified in addition to probability distributions at different locations, which can be described by correlation function and correlation length (or scale of fluctuation). Two commonly-used correlation functions, namely single exponential correlation function (SExp) and squared exponential correlation function (QExp), are used in NIGPA. The correlation functions for 2-D RFM can be written as,

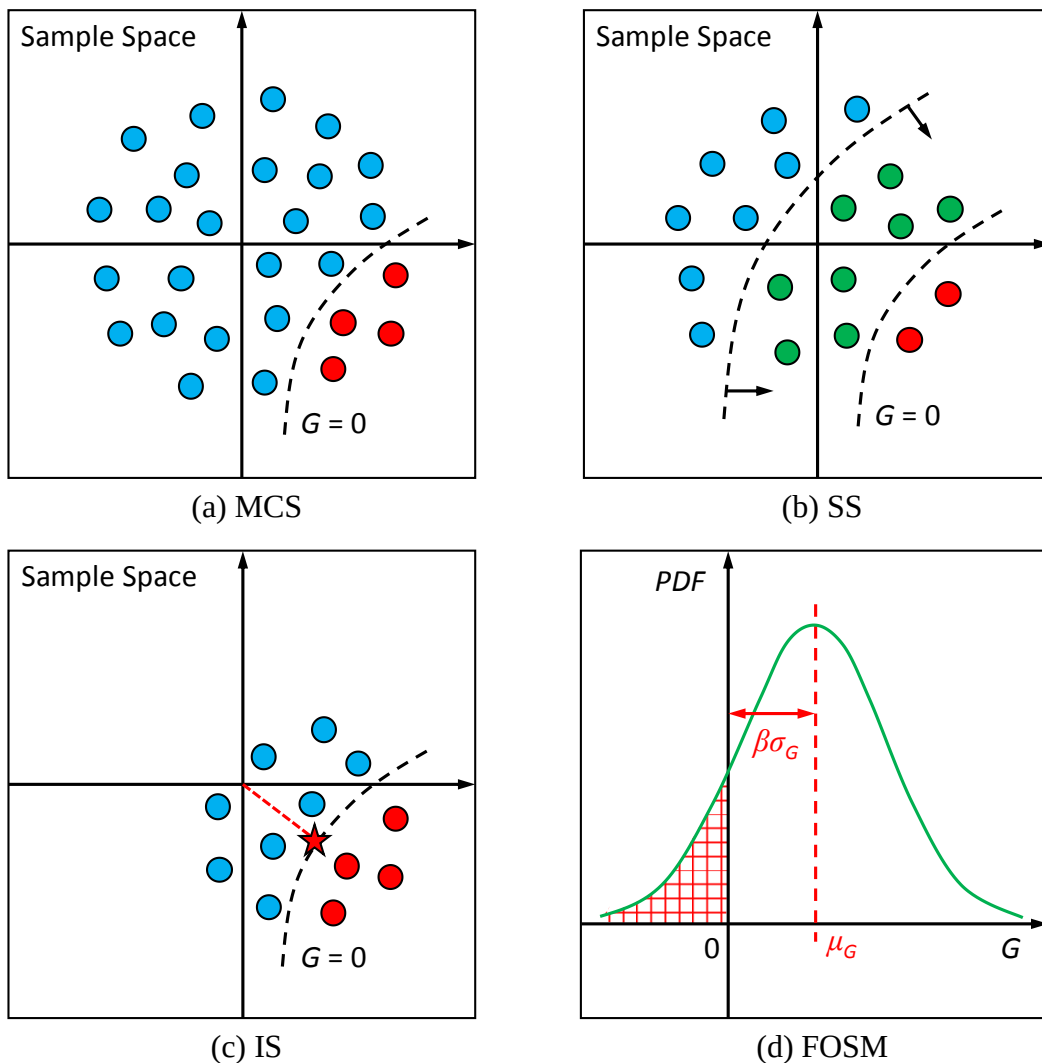
$$\text{SExp: } \rho = \exp\left(-\frac{2|\Delta_h|}{\delta_h} - \frac{2|\Delta_v|}{\delta_v}\right) \quad (1)$$

$$\text{QExp: } \rho = \exp\left[-\pi\left(\frac{\Delta_h}{\delta_h}\right)^2 - \pi\left(\frac{\Delta_v}{\delta_v}\right)^2\right] \quad (2)$$

where  $\rho$  is the correlation coefficient;  $\Delta_h$  and  $\Delta_v$  are horizontal and vertical distances, respectively, between two locations;  $\delta_h$  and  $\delta_v$  are horizontal and vertical correlation lengths, respectively. Physically,  $\delta_h \geq \delta_v \geq 0$ . If  $\delta_h$  approaches infinity, the uncertainty model turns to be 1-D RFM; if both  $\delta_h$  and  $\delta_v$  approach infinity, it turns to be RVM. With a given correlation function, the mid-point method with correlation matrix decomposition technique (Li et al., 2016c) is adopted in NIGPA to generate the random fields of soil properties. Specifically, Cholesky decomposition and Eigen-decomposition are used for SExp and QExp correlation functions, respectively.

## 2.4 Uncertainty Propagation

Several commonly-used probabilistic analysis approaches are provided by NIGPA for uncertainty propagation and reliability analysis, including Monte Carlo simulation (MCS), subset simulation (SS), importance sampling method (IS), first-order second-moment method (FOSM), first-order reliability method (FORM), and response surface method (RSM), as shown in Figure 2.4 schematically. Among these, MCS and SS are available for both RVM and RFM due to their flexibility in high-dimensional reliability problems. The rest approaches are recommended to be used only when RVM is applied in NIGPA.





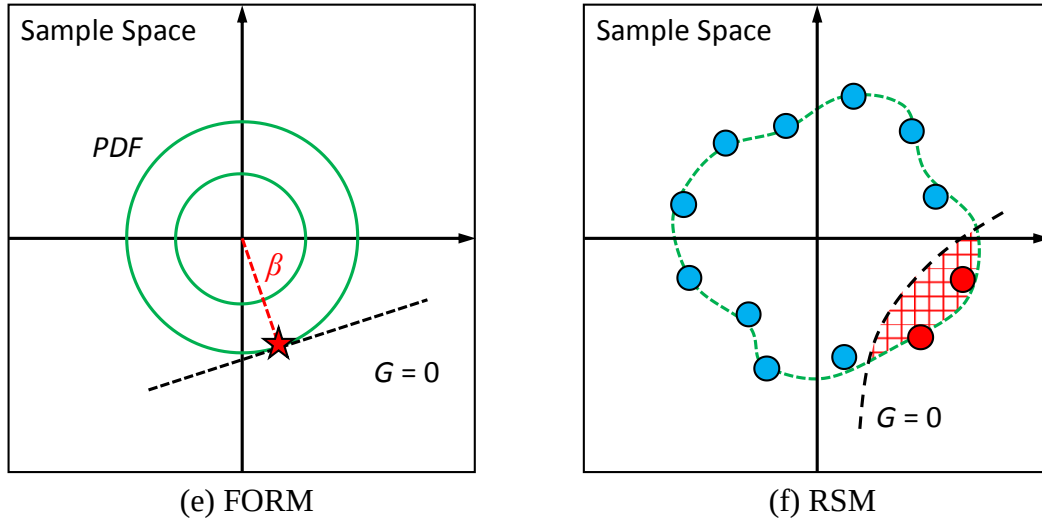


Figure 2.4 Schematic diagram for different uncertainty propagation methods

Through the above reliability analysis methods, uncertainties of input variables are propagated into responses of geotechnical structures. Let  $\mathbf{X} = [X_1, X_2, \dots, X_n]$  be a set of input random variables and  $G = g(\mathbf{X})$  be the preformation function determined by response and safety requirement. The major purpose of reliability analysis is to evaluate the failure probability  $P_f$ , which is defined as the probability that  $G$  is smaller than 0, i.e.,  $P_f = P[g(\mathbf{X}) < 0]$ .

#### 2.4.1 Monte Carlo Simulation (MCS)

In the context of MCS (Ang and Tang, 2007),  $P_f$  can be obtained through repeated random sampling and be calculated as

$$P_f = \int I[g(\mathbf{x})] f(\mathbf{x}) d\mathbf{x} = \frac{1}{N_T} \sum_{i=1}^{N_T} I[g(\mathbf{x}_i)] \quad (3)$$

where  $I[\cdot]$  is the indicator function, and it is taken as 1 if  $g(\mathbf{X}) < 0$ ; otherwise it equals to 0;  $f(\mathbf{x})$  is the joint probability density function of  $\mathbf{x}$ ;  $\mathbf{x}_i$  is  $i$ -th random sample;  $N_T$  is the number of total samples. MCS is conceptually simple and robust, but requires a large amount of samples for small probability problem.

#### 2.4.2 Subset Simulation (SS)

SS (Au and Wang, 2014) is an advanced MCS aiming for small probability problem. It stems from the idea that a small failure probability can be expressed as a product of larger conditional failure probabilities for some intermediate failure events, thereby converting a rare event simulation problem into a sequence of more frequent ones. Applying SS with  $m$  simulation levels,  $P_f$  can be estimated as

$$P_f = P(F_m) = P(F_1) \prod_{k=2}^m P(F_k | F_{k-1}) \quad (4)$$

where  $F_k = \{G < g_k, k = 1, 2, \dots, m\}$ ;  $g_1 > g_2 > \dots > g_m = 0$  is a decreasing sequence of intermediate threshold values, which can be automatically determined during SS;  $P(F_1)$  and  $P(F_k|F_{k-1})$ ,  $k = 2, 3, \dots, m-1$ , always correspond to a common specified value of conditional probability  $p_0$ ;  $P(F_m|F_{m-1})$  can be estimated based on the conditional samples in the last simulation level. Let  $N$  be the sample size in each simulation level. The total sample size for SS is  $N+mN(1-p_0)$ .

### 2.4.3 Importance Sampling Method (IS)

IS (Ching et al., 2009) is another advanced MCS. The key idea of IS is to generate samples having a higher rate of falling in the failure region. By this means,  $P_f$  in Equation (3) can be rewritten as,

$$P_f = \int I[g(\mathbf{x})] \frac{f(\mathbf{x})}{p(\mathbf{x})} p(\mathbf{x}) d\mathbf{x} = \frac{1}{N_T} \sum_{i=1}^{N_T} I[g(\mathbf{x}_i)] \frac{f(\mathbf{x}_i)}{p(\mathbf{x}_i)} \quad (5)$$

where  $p(\mathbf{x})$  is the IS density function introduced to reduce the estimation variance. Generally, it is not easy to choose a proper  $p(\mathbf{x})$ . For simplicity, the probability density function with mean located at design point and unchanged covariance is used. The design point is found using FORM, as discussed later.

### 2.4.4 First-Order Second-Moment Method (FOSM)

FOSM (Christian et al., 1994) is a simplified reliability analysis method. It aims to linearize the performance function at mean value of  $\mathbf{X}$  using Taylor series and evaluates reliability index  $\beta$  as

$$\beta = \frac{\mu_G}{\sigma_G} = \frac{g(\boldsymbol{\mu}_X)}{\sqrt{\mathbf{G}_X^T \mathbf{C}_X^{-1} \mathbf{G}_X}} \quad (6)$$

where  $\mu_G$  and  $\sigma_G$  are mean and standard deviation of  $G$ , respectively;  $\boldsymbol{\mu}_X$  and  $\mathbf{C}_X$  are mean vector and covariance matrix of  $\mathbf{X}$ , respectively;  $\mathbf{G}_X$  is the gradient vector at  $\boldsymbol{\mu}_X$ , and it can be estimated by central finite difference with interval  $k \times \boldsymbol{\sigma}_X$ ;  $\boldsymbol{\sigma}_X$  is standard deviation vector of  $\mathbf{X}$ . Generally,  $k = 1$  is suggested. Assume  $G$  is normally distributed.  $P_f$  can be calculated as  $P_f = \Phi(-\beta)$ , where  $\Phi$  is cumulative distribution function of standard normal distribution. Applying FOSM, only  $2n+1$  times deterministic analyses are needed.

### 2.4.5 First-Order Reliability Method (FORM)

Compared with FOSM, FORM (Low, 2014) linearizes the performance function at so-called design point. Design point is referred to the point located at limit state surface (i.e.,

$g(\mathbf{x}) = 0$ ) with maximum probability. Equivalently, the reliability index  $\beta$  can be estimated using optimization technique, that is,

$$\beta = \min \left\{ \sqrt{\mathbf{u}^T \mathbf{u}} \mid g(\mathbf{x}) = 0 \right\} \quad (7)$$

where  $\mathbf{u}$  is the random sample in standard normal space. Similarly,  $P_f$  also can be approximately calculated as  $P_f = \Phi(-\beta)$ .

#### 2.4.6 Response Surface Method (RSM)

To avoid expensive direct deterministic analysis, RSM (Li et al., 2016d) can be used to make the performance function explicit. For simplicity, only second-order polynomial chaos, without cross terms, is considered to construct an approximate response  $G'$ , that is,

$$G' = a_0 + \sum_{i=1}^n a_{1i} x_i + \sum_{i=1}^n a_{2i} x_i^2 \quad (8)$$

where  $\{a_0, a_{1i}, a_{2i}, i = 1, 2, \dots, n\}$  are coefficients that need to be determined. They can be calibrated by regression analysis based on several random samples. Once Equation (8) is determined, all the other reliability analysis methods can be used to estimate  $P_f$ .

## 3 Structure

### 3.1 Interface

NIGPA has four major interfaces as shown in Figure 3.1, corresponding to four modules mentioned in previous section. The real-time feedback will be displayed in the message area during the whole analysis.

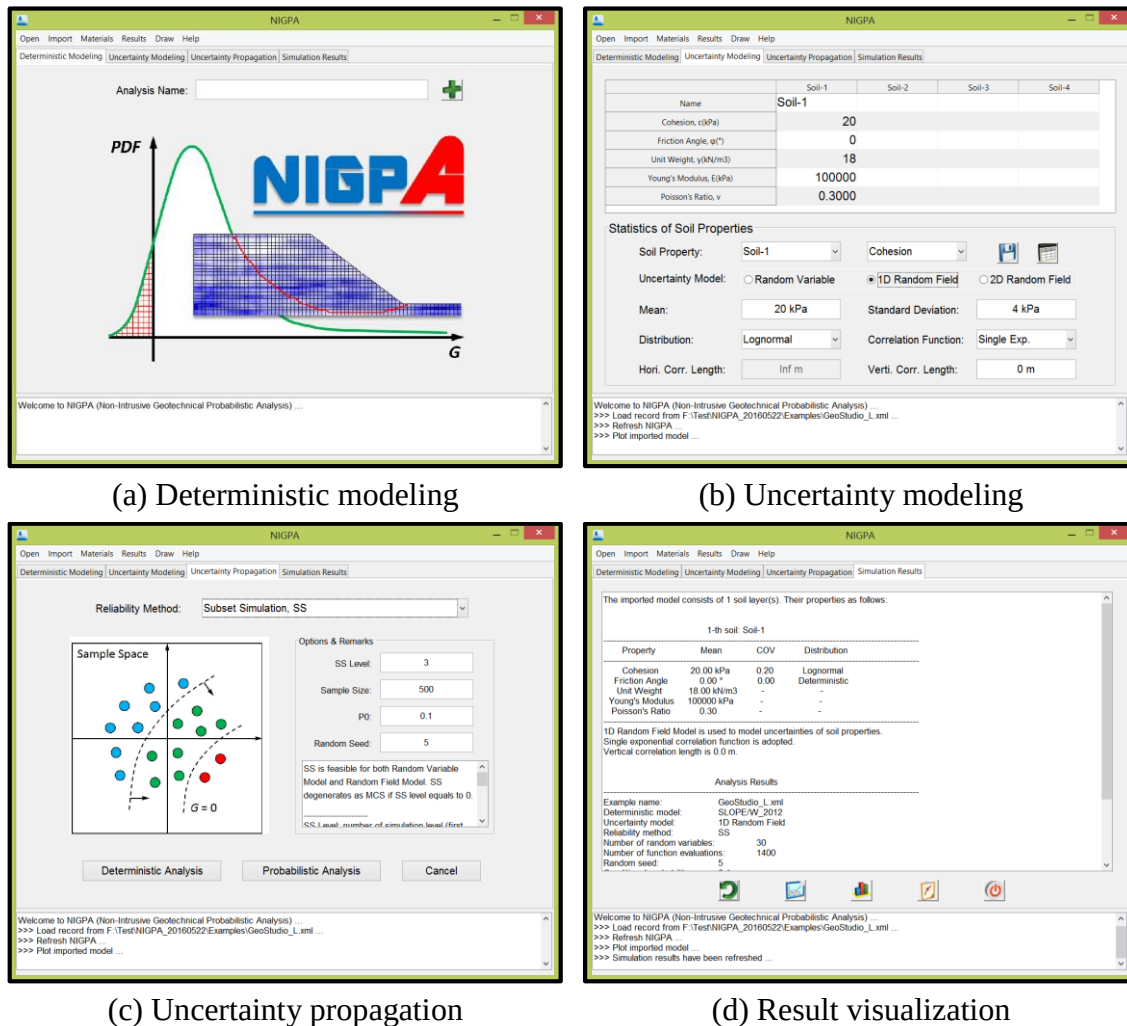


Figure 3.1 Major interfaces of NIGPA

### 3.2 File

Files contained in NIGPA are shown in Table 3.1, including 34 m-files and 4 fig-files. In addition to these, three subdirectories named “Examples”, “Help” and “Pictures” are also attached. Demonstration video and tutorial can be found in “Help” subdirectory.

Table 3.1 List of files in NIGPA

| <b>File name</b>   | <b>Function</b>                                                                    |
|--------------------|------------------------------------------------------------------------------------|
| deleteresults.m    | Delete temporary finite element analysis results                                   |
| dispAbout.m        | Provide related information about NIGPA                                            |
| dispReference.m    | Provide reference for performance level                                            |
| DrawResult.m       | GUI for simulation results                                                         |
| DrawResult.fig     | As above                                                                           |
| fileRead.m         | Read text file into “cell” data                                                    |
| fileWrite.m        | Write “cell” data into text file                                                   |
| findjob.m          | Find java objects contained within a specified java container or Matlab GUI handle |
| genbatinp.m        | Generate bat-file for inp-file                                                     |
| genbatxml.m        | Generate bat-file for xml-file                                                     |
| geninp.m           | Generate inp-file for Abaqus                                                       |
| genMatInfo.m       | Generate material information for GUI - MatInfo                                    |
| genpy.m            | Generate py-file for Abaqus                                                        |
| genxml.m           | Generate xml-file for GeoStudio                                                    |
| MatInfo.m          | GUI for summation of material properties                                           |
| MatInfo.fig        | As above                                                                           |
| NIGPA.m            | GUI for main routine of NIGPA                                                      |
| NIGPA.fig          | As above                                                                           |
| plotgeodet.m       | Plot deterministic realization for geotechnical structure                          |
| plotgeopro.m       | Plot probabilistic realization for geotechnical structure                          |
| printOut.m         | Print feedback in message area                                                     |
| Pfun.m             | Performance function for reliability analysis                                      |
| Pfun_RV.m          | Performance function for reliability analysis with random variable model           |
| readinpdata.m      | Read basic information from inp-file                                               |
| readxmldata.m      | Read basic information from xml-file                                               |
| RF_Decomposition.m | Generate random field data from geodata                                            |
| searchline.m       | Search given line in “cell” data                                                   |
| simulation_det.m   | Perform deterministic analysis                                                     |
| simulation_form.m  | Perform probabilistic analysis using FORM                                          |
| simulation_fosm.m  | Perform probabilistic analysis using FOSM                                          |
| simulation_is.m    | Perform probabilistic analysis using IS                                            |
| simulation_rsm.m   | Perform probabilistic analysis using RSM                                           |
| simulation_ss.m    | Perform probabilistic analysis using SS or MCS                                     |

| <b>File name</b> | <b>Function</b>                     |
|------------------|-------------------------------------|
| SimSettings.m    | GUI for simulation settings         |
| SimSettings.fig  | As above                            |
| SlopeAnalyze.m   | Perform slope stability analysis    |
| xml_read_yg.m    | Read xml-file into “structure” data |
| xml_write_yg.m   | Read “structure” data into xml-file |

## 4 Example

TC304 benchmark example #6 (<http://140.112.12.21/issmge/tc304.htm>) (reliability problem of slope stability) (Wang et al., 2009) is used to illustrate the uses of NIGPA, as shown in Figure 4.1. The slope example contains only one soil layer. The undrained shear strength is lognormally distributed with mean of 20kPa and coefficient of variation of 0.2. The unit weight is deterministic with a value of 18kN/m<sup>3</sup>.

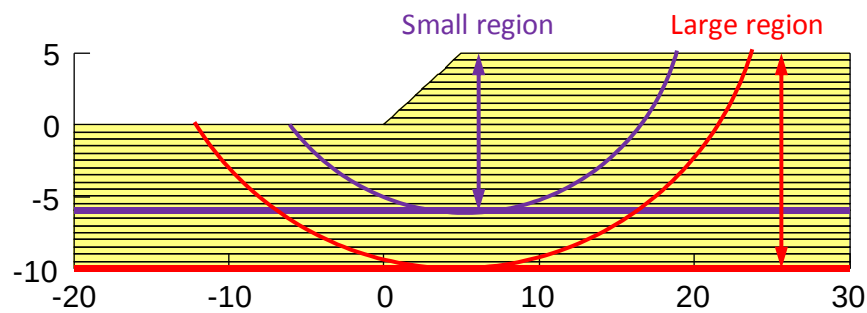
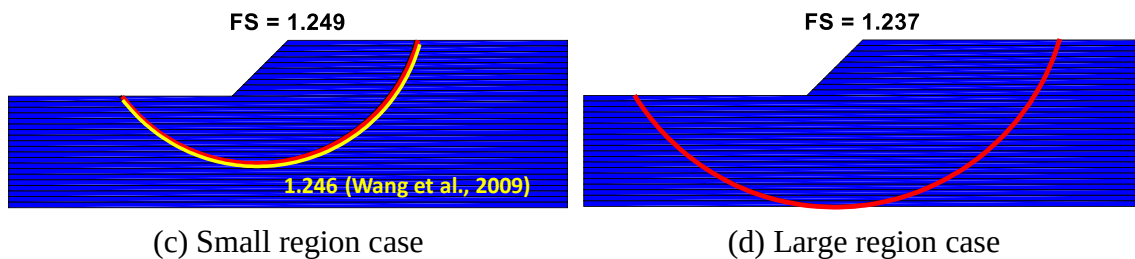


Figure 4.1 TC304 benchmark example #6 (unit: m)

The deterministic model is first developed in third-party software GeoStudio 2012 and then saved as xml-file. The reported benchmark example only searches slip surfaces in a small region with largest depth limited to -6m. For comparison, this demonstration also searches slip surfaces with largest depth up to -10m. These two cases are referred to “small region case” and “large region case”, respectively. Their corresponding models (i.e., “Slope1L\_RF1\_2012\_SmallScale.xml” and “Slope1L\_RF1\_2012\_LargeScale.xml”) can be found in “Examples” subdirectory. Based on the mean value of soil properties, the deterministic analysis results are shown in Figure 4.2. The “large region case” has a larger slip surface and a smaller factor of safety. The two models are then imported into NIGPA to perform probabilistic analysis. Figure 4.3 gives an example for the “large region case”.



(c) Small region case

(d) Large region case

Figure 4.2 Deterministic analysis results of benchmark example

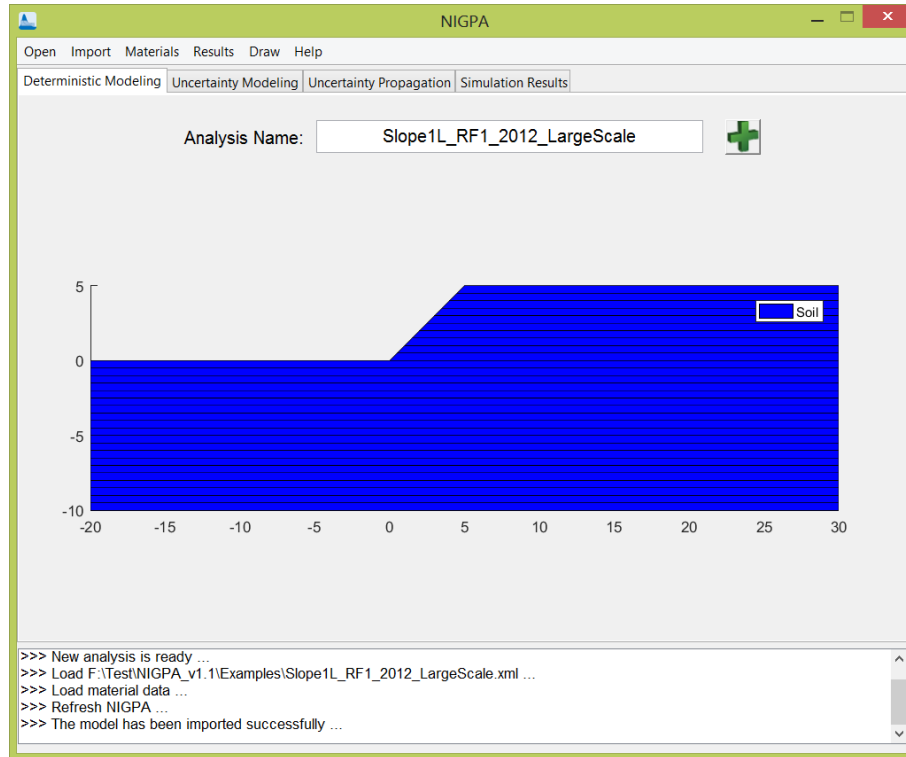


Figure 4.3 Example for importing deterministic model into NIGPA

In the benchmark example, the soil layer is divided into 30 independent and identically distributed layers, which is a special case of random field model with perfect correlation in horizontal direction and full independence in vertical direction (i.e., 1-D random field model with  $\delta_v = 0\text{m}$ ). Applying this uncertainty model, several analyses are carried out using NIGPA, including one MCS run with  $N_T = 10,000$  and five SS runs with  $N = 500$ ,  $m = 2$  and  $p_0 = 0.1$ . Table 4.1 compares the probabilistic analysis results of the benchmark example. The results obtained from NIGPA agree well with the reported results (Wang et al., 2009). The failure probability of “large region case” is almost three times larger than that of “small region case”.

Table 4.1 Probabilistic analysis results of benchmark example

| Case         | Method | Wang et al. (2009)   | NIGPA                |
|--------------|--------|----------------------|----------------------|
| Small region | MCS    | $1.6 \times 10^{-3}$ | $1.4 \times 10^{-3}$ |
|              | SS     | $1.2 \times 10^{-3}$ | $1.4 \times 10^{-3}$ |
| Large region | MCS    | —                    | $3.8 \times 10^{-3}$ |
|              | SS     | —                    | $3.5 \times 10^{-3}$ |



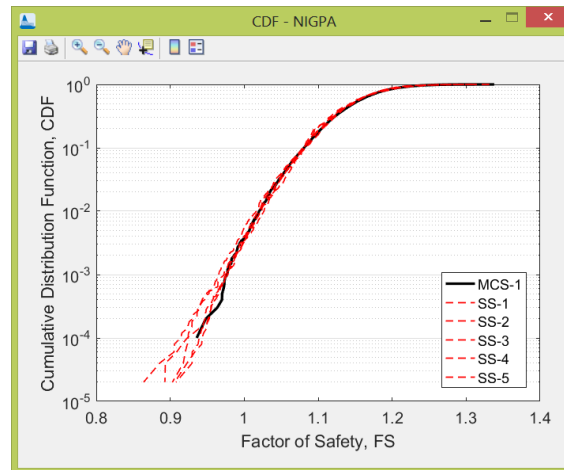


Figure 4.4 Comparison for cumulative distribution function of response

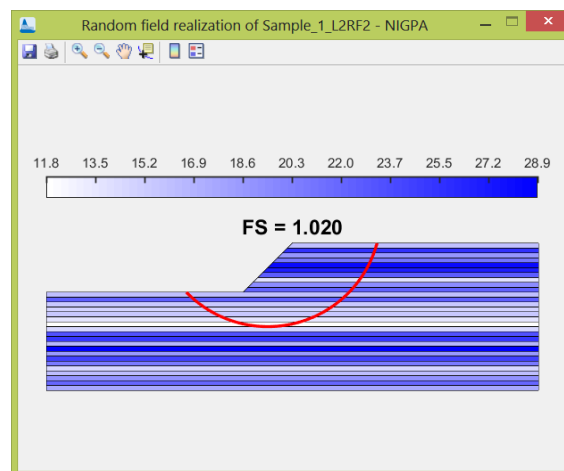


Figure 4.5 View of random field realization and corresponding results

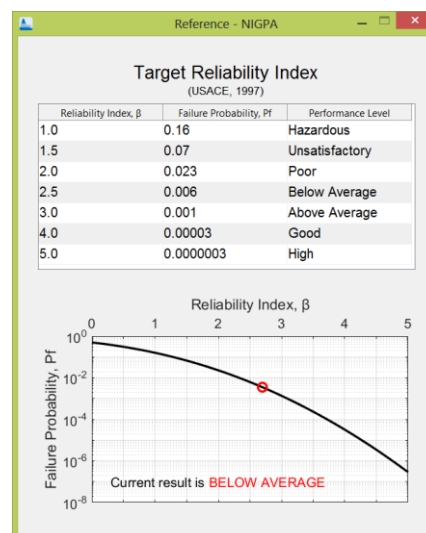


Figure 4.6 Assessment of the performance level

In addition to failure probability estimation, more reliability analysis results can be obtained from NIGPA, such as comparison for cumulative distribution function of response obtained from different analyses (see Figure 4.4), view of random field realization and corresponding results (see Figure 4.5), and assessment of the performance level of geotechnical structures (USACE, 1997) (see Figure 4.6). More details can be referred to the demonstration video.

In conclusion, users only need to focus on developing a deterministic analysis model in commercial software, and the geotechnical probabilistic analysis can be achieved with relatively ease using NIGPA.

## 5 Additional Notes

To avoid unexpected errors in NIGPA, the deterministic model produced by third-party commercial software must be able to perform stability analysis successfully to obtain the corresponding factor of safety. More particularly,

- (1) For limit equilibrium model developed by GeoStudio,
  - (a) Slope/W should be set as the major module;
  - (b) Path of “GeoStudio.exe” should be correctly set in NIGPA, before probabilistic analysis;
  - (c) If random field model will be used, the random field mesh, corresponding to “Regions”, should be partitioned in GeoStudio. More details can be referred to the attached examples. Specially, the number of random field mesh is highly recommended to be smaller than 1000; otherwise, the probabilistic analysis may be interrupted by “out of memory” warning given by GeoStudio.
- (2) For finite element model developed by Abaqus,
  - (a) Only plane strain element “CPE” is supported for finite element mesh;
  - (b) Only Mohr-Coulomb failure criterion is supported for soil property;
  - (c) Soil weight should be modeled as “body force”;
  - (d) Shear strength reduction procedure should be achieved successfully. The keywords for “field variable” should start from “\*Initial conditions, type=field, variable=1” and “\*Field, variable=1”. The loading step should be named as “Load” with 1s time period, and the shear strength reduction step should be named as “Reduce” with 4.5s time period. More details can be referred to the attached examples. Specially, if the actual factor of safety is smaller than 0.5 or larger than 5.0, the results will be erratic since default shear strength reduction factor ranges from 0.5 to 5.0.

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